PRE- BOARD EXAMINATION (2024-25)

CLASS – X

TIME: 3 Hrs SUBJECT: MATHEMATICS STANDARD (041) M.M: 80

MARKING SCHEME

S. No.	SOLUTION	MARKS
	SECTION A	
1	(b)$2\sqrt{a^2-b^2}$	1
2	(c)-3/2	1
3	(b) 4	1
4	(b)2:3	1
5	(b) (0,-1)	1
6	(b) (7,0)	1
7	(c)1	1
8	(c) 400	1
9	(b) 7.5 cm	1
10	(a)40	1
11	(d) $\sqrt{119}$	1
12	-3	1
13	(b) 1/12	1
14	(d) 150^0	1
15	(a)120^0	1
16	45^0	1
17	(a)1/6	1
18.	(c) $\sqrt{(b^2-a^2)}/b$	1
19	(b)	1
20	(c)	1
	SECTION B [2 X 5 =10]	
21.	Let $2-3\sqrt{5}$ be rational $2-3\sqrt{5} = p/q$, $q \neq 0$, p and q are coprimes $3\sqrt{5}= p/q -2$ $\sqrt{5} = \frac{p-2q}{3q}$ Which means $\sqrt{5}$ is also rational as RHS is rational But this contradicts the fact that $\sqrt{5}$ is irrational . So our assumption is wrong. Hence $2-3\sqrt{5}$ is irrational.	½ M 1M ½ M
22.	For correct proof…………	2 M

23	Here, AO = BO $\angle$ OBA = $\angle$ OAB $\angle$ OBA + $\angle$ OAB + $\angle$AOB = 180 degree (Angle Sum property) From above, 2($\angle$ OAB) = 80^o $\angle$ BAT = 50^o OR Let $\angle PTQ = \theta$ $\angle TPQ = \angle TQP = ½(180^o - \theta) = 90 - \theta/2$ $\angle OPQ = 90^o – (90^o - \theta/)2 = \theta/2 = ½ \angle PTQ$ $\angle PTQ = 2\angle POQ$	½ M 1M 1/2 M 1M 1M
24	Area of sector = $\theta/360$ X πr^2 $\theta/360$ X 22/7 X 1225 = 3850/4 $\theta = 90^o$ **Or** Angle swept in 5 mins = 30^o Area of sector = $\theta/360$ X πr^2 30/360 X 22/7 X r^2 = 154/3 $r^{2 =}$ 196 , r = 14 cm	1/2 M 1M 1/2 M 1/2 M 1/2 M 1M
25.	For Correct fig. $\tan\theta = 20\sqrt{3}/60 = 1/\sqrt{3}$ $\tan\theta = \tan 30$, $\theta = 30$ degree	1M 1M
	SECTION C [3 X 6 = 18]	
26	Let us assume to the contrary that ($\sqrt{5}$- $\sqrt{3}$) be a rational number ($\sqrt{5}$- $\sqrt{3}$) = a where a is a rational number Squaring both sides (Using$(a+b)^2=a^2+b^2+2ab$) $8 - 2\sqrt{15} = a^2$ $\sqrt{15} = (8 - a^2)/2$ Irrational = rational Contradiction due to incorrect assumption ($\sqrt{5}$- $\sqrt{3}$) is irrational.	1M 1M 1M
27.	$\alpha + \beta = -b/a = 7/6$ $\alpha\beta = c/a = 2/6 = 1/3$ $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $\alpha^2 + \beta^2 = 25/36$	 1M 1M 1M
28.	$a^2x + b^2y = c^2$ (1) $b^2x + a^2y = d^2$ (2) multiply eq 1 by a^2 and eq 2 by b^2 and eliminating $x = (a^2c^2 - b^2d^2)/a^4 – b^4$ and $y = (a^2d^2 – b^2c^2)/a^4 - b^4$	 1M 2M

<table>
<tr><td></td><td>OR
434x + 262 y = 1826 -------(1)
262x + 434y = 1654 --------(2)
Adding eqn. 1 and 2
696x + 696 y = 3480. This implies 696 (x + y) = 3480.
Hence x + y = 5 –(3)
Eqn. 1 – Eqn. 2
172 x – 172 y = 172 Hence x – y = 1 ------(4)
Solving eqns. 3 and 4
2x = 6 so x = 3
And 2y = 4 so y = 2</td><td>1M

1M

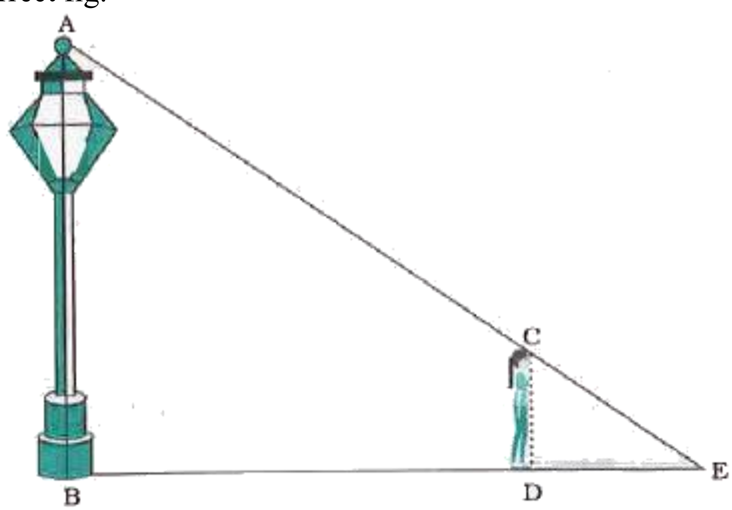
1M</td></tr>
<tr><td>29</td><td>For correct proof…….</td><td>3 M</td></tr>
<tr><td>30</td><td>For correct proof…….
Or
ΔOAT ≡ ΔOBT
$\angle$ ATB = 2$\angle$ ATO = 80^{o}
$\angle$ AOB= 180^{o} – 80^{o} = 100^{o}</td><td>3M

1M
1M
1M</td></tr>
<tr><td>31</td><td><table>
<tr><td>CI</td><td>F</td><td>CF</td></tr>
<tr><td>0-6</td><td>4</td><td>4</td></tr>
<tr><td>6-12</td><td>X</td><td>4 + x</td></tr>
<tr><td>12-18</td><td>5</td><td>9+ x</td></tr>
<tr><td>18-24</td><td>Y</td><td>9 + x + y</td></tr>
<tr><td>24-30</td><td>1</td><td>10 + x + y</td></tr>
<tr><td></td><td></td><td></td></tr>
</table>
10 + x + y = 20, x + y = 10
Median = $1 + \frac{\frac{n}{2} - C}{f}$ X h
$14.4 = 12 + \frac{10-4-6}{5}$ X 6
Solving further
x = 4 , y = 6</td><td>1M for correct table

1M

1M</td></tr>
<tr><td colspan="3">SECTION D [5 X 4 =20]</td></tr>
<tr><td>32</td><td>For equal roots $b^2 – 4ac = 0$
a= k+1   b = -6k-2   c = 8k + 1
36 k^2 + 4 + 24k – 4 (k+1)(8k +1) = 0
Solving further
$K^2 – 3k = 0$
K (k-3) = 0</td><td>1M

3M</td></tr>
</table>

	K = 0 or k = 3 Or **let** the number of articles be x, then cost of each article = 2x + 3 ATQ $x(2x+3) = 90$ or $2x^2 + 3x - 90 = 0$ Solving the equation x = -15/2 (rejected) or x = 6 (correct) Number of articles = 6 cost of each article = Rs. 15	1M 1M 2M 2M
33	. For the Theorem : statement 1M Correct Proof In triangle KMN , PQ // MN 4/13 = KQ/ KN-KQ 4/13 = KQ/ 20.4-KQ 17 KQ = 4 X 20.4 KQ = 4.8 cm OR Correct fig.  Let DE = x m, BD= 1.2 X 4 = 4.8 m $\Delta ABE \sim \Delta CDE$ (AA) BE/DE = AB/CD (4.8 + x)/x = 3.6/0.9 X = 1.6 m	1M 3M 1M 1M 2M 2M
34	r = 3.5 m, h = 3 m L= 10m, B= 7 m interior surface of the building= area of four walls + TSA of half cylinder $= 2(l+b)h + \pi r^2 + \pi rh$ (Substituting values and correct calculation) $=250.5\ m^2$	1M 1M 3M

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1. Let the assumed mean A be 145.

Class interval h = 10.

Class	Frequency fi	Midvalue xi	Fixi
20-25	7	22.5	157.5
25-30	6	27.5	165
30-35	9	32.5	292.5
35-40	13	37.5	487.5
40-45	F	42.5	42.5f
45-50	5	47.5	237.5
50-55	4	52.5	210

Σ Fi= 44+ f Σ fixi = 1550 + 42.5f

For **correct table** 2M

(i) Mean $\bar{x} = \Sigma$ fixi/ Σ Fi 1M

Substituting values and solving for f 2M

f = 6

SECTION E [4 X 3 =12]

36.	i) Distance between Neena and Karan= $\sqrt{10}$ unit ii) (2,3) iii) point = (7/2, 5/2) or mid point=(4,4)	1M 1M 2M
37	Yes it is an A.P. first term=240, common difference = 60 (ii) 240 + (n-1) 60 = 660 n = 8 saving will be 660 in 8th month. (iii) a_{15} = a + 14 d = 240 + 14 X 60= 1080 OR S_{10} = 5(2 X 240 + 9 X 60)= 2040	1M 1M 2M
38	(i) BD=6m (ii) 12/$\sqrt{3}$ Or DC=6/$\sqrt{3}$ (iii) Angle is 45^0	1M 2M 1M